Residual Physics-Informed Neural Networks for Joint Parameter Identification in Belt-Driven Robotic Systems

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Abstract. Accurate identification of dynamic parameters in robotic joints is essential for model-based control. Traditional methods struggle with belt-driven transmission systems where friction and compliance effects dominate dynamics and violate standard modeling assumptions. This paper presents a residual Physics-Informed Neural Network (PINN) architecture that addresses a critical limitation of standard PINNs: the tendency to memorize input-output mappings rather than learning physically meaningful parameters. Our architecture explicitly decomposes measured torque into a physics-based component (with learnable inertia and damping parameters) and a neural network residual capturing unmodeled dynamics. After validating the method on a simulated PMSM motor system we validate the approach on the Solo-12 quadruped robot's hip flexion/extension joints using multiple datasets with varying sampling rates. The method provides interpretable parameter estimates directly applicable to model-based control while offering insights into the dominant dynamics of belt-driven transmissions.

Keywords: Physics-Informed Neural Networks, Parameter Identification, Robot Dynamics, Belt-Driven Transmission, Solo Robot

1 Introduction

Accurate knowledge of dynamic parameters such as inertia and damping is fundamental for model-based control strategies in robotics. Inertia determines feedforward compensation for acceleration, while damping knowledge enables stable gain tuning and accurate trajectory tracking. While CAD models provide nominal values, real-world robots exhibit significant deviations due to manufacturing tolerances, assembly variations, and unmodeled effects [1]. This discrepancy is particularly pronounced in belt-driven transmission systems. Belt drives are common alternatives to chain and gear transmissions in automotive and industrial engineering. They offer several key advantages, including simplified installation, low maintenance needs, and effective shock absorption [2].

Belt-driven transmissions present unique identification challenges compared to rigid gear trains. Belt drives exhibit significant compliance, velocity-dependent friction, and temperature sensitive behavior. These characteristics violate the linear-in-parameters assumption underlying classical identification methods, where dynamic equations can be expressed as $\mathbf{Y}(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})\boldsymbol{\theta} = \boldsymbol{\tau}$. Success critically depends on accurate derivative estimation from position measurements, which becomes problematic with noisy data or limited sampling rates [3].

Physics-Informed Neural Networks (PINNs) [4] offer an alternative by encoding physical laws as soft constraints in the loss function, enabling data-driven learning while respecting governing equations. Extensions to mechanical systems [5] have shown promise for parameter estimation. Recent work has further extended PINNs to robotic parameter identification: Yang et al. [10] applied PINNs to model predict and identify dynamic parameters of collaborative robot joints, embedding rigid-body dynamics directly into the learning framework. In the domain of belt-driven systems, Baqer et al. [11] demonstrated that neural network approaches combined with vibration analysis can effectively predict belt drive condition, highlighting the relevance of data-driven methods for transmission systems. However, standard PINNs struggle with the high-frequency vibrations inherent in belt-driven systems. Instead of identifying true physical parameters, the network often compensates by adjusting its weights to fit the data points [6].

This paper presents a Residual PINN [7] architecture that addresses this problem by explicitly decomposing measured torque into interpretable physical components with learnable inertia and damping parameters, while a neural network residual captures unmodeled dynamics. We validate the approach on the Solo-12 quadruped robot's hip joints and benchmark it against classical least-squares identification adapted from [9], demonstrating robust parameter identification across varying excitation conditions.

The remainder of this paper is organized as follows: Section 2 presents the residual PINN methodology. Section 3 describes the experimental setup including PMSM simulation validation and the Solo-12 platform. Section 4 presents identification results. Section 5 concludes with key findings and future directions.

2 Methodology

2.1 Problem Formulation

Consider a single robotic joint with the following dynamics:

$$\tau = J\dot{\omega} + b\omega + \tau_{\text{ext}} \tag{1}$$

where τ is the joint-side torque, J is the effective joint-side inertia, $\dot{\omega}$ is the angular acceleration, b is the viscous damping coefficient, ω is the angular velocity, and τ_{ext} represents external and unmodeled torques. The goal is to identify J and b from measurements of τ, ω, and time t.

When learnable physical parameters are embedded directly in a neural network and trained end-to-end, the optimization has no incentive to find physically meaningful values. The network can compensate for arbitrary parameter choices by adjusting its weights, treating inertia and damping as free variables rather than physical constants. To address this, we propose a residual architecture that structurally separates physics from learned components.

2.2 Residual PINN Architecture

A Residual PINN is a deep feedforward architecture that utilizes skip connections to bridge its layers. This design allows each layer's output to be a combination of its own nonlinear transformation and the direct input from the previous stage, [7]. We reformulate the problem by explicitly decomposing the measured torque:

$$\tau_{\text{measured}} = \underbrace{J\alpha + b\omega}_{\tau_{\text{physics}}} + \underbrace{\mathcal{N}_\theta(t, \omega, \alpha)}_{\tau_{\text{residual}}} \tag{2}$$

where $\alpha = \dot{\omega}$ is computed via first-order finite differences, J and b are learnable scalar parameters, and $\mathcal{N}_\theta$ is a neural network learning only the residual torque not explained by the physics model.

The residual network takes three inputs (normalized time, velocity, and acceleration) and produces a single output through two hidden layers of 64 neurons with Tanh activation. This architecture was selected to balance expressiveness with robust parameter estimation, as wider and deeper networks caused the residual to absorb physically meaningful torque components, thereby degrading the estimation of J and b. Tanh was preferred over ReLU for smoothness, which is better suited to continuous mechanical dynamics. Xavier uniform initialization with a small gain (0.01) ensures near-zero initial residuals, forcing the physics model to explain the data first.

The combined loss function is:

$$\mathcal{L} = \mathcal{L}_{\text{physics}} + \lambda \mathcal{L}_{\text{residual}} \tag{3}$$

where:

$$\mathcal{L}_{\text{physics}} = \frac{1}{N} \sum_{i=1}^{N} \left(\tau_i - J\alpha_i - b\omega_i - \mathcal{N}_\theta(t_i, \omega_i, \alpha_i)\right)^2 \tag{4}$$

$$\mathcal{L}_{\text{residual}} = \frac{1}{N} \sum_{i=1}^{N} \mathcal{N}_\theta(t_i, \omega_i, \alpha_i)^2 \tag{5}$$

The residual regularization term ($\lambda = 0.1$) encourages the network to keep residuals small, forcing the physics model to explain as much torque as possible.

3 Experimental Setup

We validated the residual PINN architecture in two stages. First, we tested on a simulated Permanent Magnet Synchronous Motor (PMSM) motor in MATLAB/Simulink, providing a controlled environment with known ground-truth

parameters to verify the method's ability to recover physical quantities. Second, we applied the approach to real hardware using the Solo-12 quadruped robot, where ground-truth parameters are unknown and belt-driven transmission dynamics introduce additional complexity.

3.1 PMSM Motor Simulation

The PMSM simulation comprised a field-oriented control (FOC) system in MATLAB/Simulink with a combined rotor and load inertia of $J = 0.28$ kg m^2 and viscous friction $B = 0.001$ N m s/rad. The motor tracked a varying speed reference over 5 s, sampled at $T_s = 1$ ms. This noise-free dataset provided an ideal testbed for validating the method before real hardware.

3.2 Solo-12 Robot Platform

The Solo-12 is an open-source quadruped robot developed by the Open Dynamic Robot Initiative [8]. We focus on the Hip Flexion/Extension (HFE) joints, which operate approximately parallel to the ground, minimizing gravitational effects.



Fig. 1. Experiment setup

Each joint is driven by a T-Motor Antigravity 4004 300kV brushless motor (53g) through a dual-stage timing belt transmission with a total gear ratio of $N = 9$ (3:1 per stage), using Conti Synchroflex AT3 GEN III belts. Position feedback is provided by a Broadcom AEDT-9810-Z00 encoder with 20,000 counts per revolution mounted on the joint side. The ODRI firmware reports all quantities, positions, velocities, and torques in joint-side units, with torques computed as $\tau_{\text{joint}} = i_{\text{motor}} \cdot N \cdot K_t$. All identified parameters therefore correspond directly to the joint side.

3.3 Datasets

We collected four datasets with the robot leg held parallel to the ground to eliminate gravitational effects. The excitation signal was a rectangular (square wave) torque command alternating between $+\tau_{\text{amp}}$ and $-\tau_{\text{amp}}$, with each half-period lasting $N_s/2$ samples at 1 kHz sampling rate. This pattern was repeated for 10 full cycles, yielding a total of $10 \times N_s$ samples per dataset. Higher torque amplitudes required shorter periods to prevent the leg from exceeding its joint limits. Data was collected from both the front-left (FL) and front-right (FR) HFE joints to assess symmetry between nominally identical joints.

Table 1. Dataset characteristics. All datasets sampled at 1 kHz with 10 full square-wave cycles.

Dataset	τ_{amp} (N m)	Period N_s (samples)	Total samples	Duration (s)
data_500_0.2	0.20	500	5000	5.0
data_800_0.15	0.15	800	8000	8.0
data_1000_0.12	0.12	1000	10000	10.0
data_1500_0.1	0.10	1500	15000	15.0

Each dataset contains synchronized measurements of joint positions, velocities, commanded torques, and measured torques for all 12 joints. Training was performed for 10000 epochs on an NVIDIA RTX 4060 GPU.

4 Results

4.1 PMSM Motor Validation Results

The validation on the simulated PMSM demonstrated excellent accuracy. Starting from $J = 0.01$ kg m^2, the residual PINN converged to $J = 0.2817$ kg m^2 after 25,000 epochs (true value: $J = 0.28$ kg m^2, relative error 0.6%). This confirmed the method's ability to recover physical parameters from dynamic data and provided confidence to proceed with real hardware.

4.2 Solo-12 Robot: Single Dataset Analysis

Table 2 presents results from the dataset that yielded the best overall performance (800 samples, 0.15 N m torque amplitude):

The physics model (inertia + damping) explains approximately 80% of the measured torque, with the neural network residual capturing unmodeled effects such as nonlinear friction, belt compliance, and Coulomb friction.

Analysis of individual torque contributions reveals that belt-driven joints are damping-dominated: the viscous damping term ($b\omega$) accounts for 55-65% of the measured torque, while the inertial contribution ($J\alpha$) represents only 15-25%,

Table 2. Identified parameters for HFE joints (data_800_0.15). J in $\mathrm{g\,m^2}$, b in $\mathrm{N\,m\,s/rad}$.

Joint	J	b	Physics (%)	Residual (%)
FL_HFE	0.181	0.106	78.3	21.7
FR_HFE	0.269	0.111	82.0	18.0
Mean	0.225	0.108	80.1	19.9

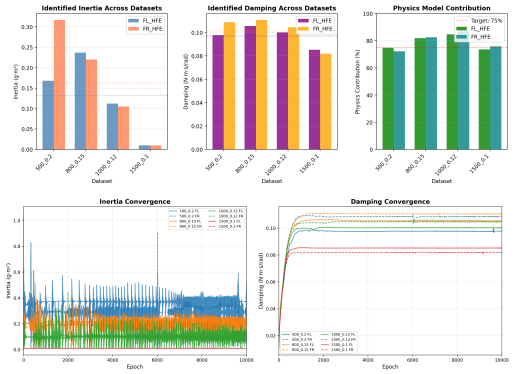


Fig. 2. Cross-dataset validation showing identified inertia and damping parameters across four datasets with varying sampling rates for 10000 epochs.

with implications for control design where accurate damping compensation is more critical than precise inertia knowledge.

Figure 2 and Table 3 present results across all four datasets: The damping coefficient shows exceptional consistency, $b = 0.099 \pm 0.010\,\mathrm{N\,m\,s/rad}$, with a Coefficient of Variation (CV) of 10%. Table 3 compares the residual PINN against classical least-squares identification using a three-parameter model $\tau = J\alpha + b\omega + F_s\,\mathrm{sgn}(\omega)$, adapted from [9] by omitting the gravitational terms since the leg operates parallel to the ground. The LS method yields lower viscous damping ($b \approx 0.014\ \mathrm{N\,m\,s/rad}$, CV = 32.2%) by separating Coulomb friction into an explicit F_s term, confirming that the PINN's higher $b \approx 0.099$ absorbs both effects. Extending the PINN physics component with a $F_s\,\mathrm{sgn}(\omega)$ term would likely decouple these contributions.

This robustness validates our method and confirms that damping is the dominant identifiable parameter in belt-driven systems. Statistical analysis shows no significant difference between nominally identical left and right joints: the mean damping values ($b_{\mathrm{FL}} = 0.097$ vs. $b_{\mathrm{FR}} = 0.102\ \mathrm{N\,m\,s/rad}$) are within one standard deviation, supporting the physical consistency of our estimates.

Inertia estimates show a strong dependence on excitation level (CV = 70.5%), with two distinct mechanisms: higher torque amplitude (0.2 N m) provides sufficient excitation for inertia identification, while low torque (0.1 N m) causes

Table 3. Cross-validation: residual PINN vs least-squares [9]. J in $\mathrm{g\,m^2}$, b in $\mathrm{N\,m\,s/rad}$. All values joint-side. CV denotes the Coefficient of Variation.

	Residual PINN				Least-Squares			
Dataset	J_{FL}	J_{FR}	b_{FL}	b_{FR}	J_{FL}	J_{FR}	b_{FL}	b_{FR}
data_500_0.2	0.284	0.258	0.098	0.109	0.373	0.290	0.013	0.018
data_800_0.15	0.181	0.269	0.106	0.111	0.213	0.190	0.013	0.019
data_1000_0.12	0.101	0.090	0.100	0.105	0.106	0.098	0.013	0.021
data_1500_0.1	0.010	0.010	0.085	0.082	−0.028	−0.040	0.006	0.011
Mean ± std	0.150 ± 0.106		0.099 ± 0.010		0.150 ± 0.136		0.014 ± 0.005	
CV (%)	70.5		10.0		90.4		32.2	

both joints to converge to the minimum clamp value ($0.01\ \mathrm{g\,m^2}$), indicating that damping dominates and inertia becomes unidentifiable. For reliable inertia identification, torque excitation $\geq 0.12\ \mathrm{N\,m}$ is recommended.

The Solo-12 URDF [8] provides link COM inertias $J_{\mathrm{cm,upper}} = 0.411\ \mathrm{g\,m^2}$ and $J_{\mathrm{cm,lower}} = 0.120\ \mathrm{g\,m^2}$. Applying the parallel axis theorem ($J_i = J_{\mathrm{cm},i} + m_i d_i^2$) with masses and segment lengths from [8] yields a total rigid-body link inertia of $J_{\mathrm{link}} = 3.507\ \mathrm{g\,m^2}$, approximately 23× larger than the identified $J \approx 0.150\ \mathrm{g\,m^2}$. This large discrepancy is consistent with belt compliance, which decouples the link-side load from the actuator during rapid transients, so that the identified inertia primarily reflects the motor rotor and pulley inertia referred to the joint side.

5 Conclusion

We presented a residual PINN architecture for joint parameter identification that addresses the memorization problem in standard PINNs. The residual formulation prevents the network from bypassing physics by predicting torque residuals rather than velocities, eliminating identity mappings. The residual regularization term encourages the physics model to explain as much torque as possible, while small weight initialization ensures that physics dominates early in training, guiding parameters toward physically meaningful values.

Applied to the Solo-12 quadruped robot, the method achieves robust damping identification (CV = 10%) and reveals that belt-driven joints are fundamentally damping-dominated. Comparison with classical least-squares identification confirms that the PINN's damping coefficient absorbs both viscous and Coulomb friction effects while maintaining superior consistency across datasets.

A key finding is that inertia identification requires sufficient torque excitation: datasets with commanded torque $\geq 0.12\ \mathrm{N\,m}$ yielded consistent inertia estimates ($J \approx 0.1$–$0.3\ \mathrm{g\,m^2}$), while low-excitation data ($0.1\ \mathrm{N\,m}$) caused inertia to collapse to the lower bound as damping dominated the dynamics. This demonstrates a fundamental limitation of derivative-dependent identification in damping-dominated systems.

The current approach assumes single-joint dynamics without gravitational coupling, which is valid for joints operating parallel to the ground but requires extension for general configurations. Future work will leverage the identified parameters for model-based PD controller tuning, where knowledge of joint inertia and damping enables systematic gain selection rather than empirical tuning. Additionally, we plan to extend the framework with nonlinear friction models, multi-joint coupling, and online parameter adaptation for time-varying belt characteristics.

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